



Lakshya JEE 2.0 (2024)

Method of Differentiation

DPP-02

1. If $\sin y = x \cos(a + y)$, find value of $\frac{dx}{dy}$.

(1) $\frac{\cos a}{\cos^2(a + y)}$

(2) $\frac{\sin a}{\sin^2(a + y)}$

(3) $\frac{\cos a}{\sin^2(a + y)}$

(4) $\frac{\sin a}{\cos^2(a + y)}$

2. $\frac{d}{dx} \left\{ \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) \right\} =$

(1) $\sqrt{x^2 - a^2}$ (2) $\sqrt{a^2 - x^2}$

(3) $\sqrt{a - x^2}$ (4) $\sqrt{x^2 - a}$

3. If $y = \frac{\sin^{-1} x}{\sqrt{1 - x^2}}$, then $(1 - x^2) \frac{dy}{dx}$ is equal to:

(1) $x + y$ (2) $1 + xy$

(3) $1 - xy$ (4) $xy - 2$

4. If $y = \tan^{-1} \sqrt{\frac{1 + \cos x}{1 - \cos x}}$, where $0 < x < \pi$, then $\frac{dy}{dx}$

is:

(1) $-1/2$ (2) 0

(3) 1 (4) -1

5. $\frac{d}{dx} \cos^{-1} \sqrt{\cos x}$ is equal to:

(1) $\frac{1}{2} \sqrt{1 + \sec x}$

(2) $\sqrt{1 + \sec x}$

(3) $-\frac{1}{2} \sqrt{1 + \sec x}$

(4) $-\sqrt{1 + \sec x}$

6. $\frac{d}{dx} \left[\sin^2 \cot^{-1} \left\{ \sqrt{\frac{1-x}{1+x}} \right\} \right]$ is equal to:

(1) -1 (2) $1/2$

(3) $-1/2$ (4) 1

7. Let $f(x) = \begin{vmatrix} 1 & 2x & 3x^2 \\ x & x^2 & x^3 \\ 0 & 2 & 6x \end{vmatrix}$

Then find the value of $f'(1)$

(1) 0 (2) -2

(3) -6 (4) 6

8. Find $\frac{dy}{dx}$, if $y = x^{\sin x}$.

(1) $x^{\sin x} \left(\frac{\sin x}{x} - \cos x \log x \right)$

(2) $x^{\sin x} \left(\frac{\sin x}{x} + \cos x \log x \right)$

(3) $x^{\sin x} \left(\frac{\sin x}{x} + \cos x \cdot \frac{1}{x} \right)$

(4) $x^{\sin x} \left(\frac{\sin x}{x} + \sin x \log x \right)$

9. If $y = \left(1 + \frac{1}{x} \right)^x + x^{\left(1 + \frac{1}{x} \right)}$, find $\frac{dy}{dx}$ at $x = 1$

(1) $\log 2 - \frac{1}{2}$ (2) $\log 2 + 1$

(3) $\log 2 - 1$ (4) $\log 4 + 1$

10. If $y = (1 + x)(1 + x^2)(1 + x^4)(1 + x^8) \dots (1 + x^{1006})$, then the value of $y'(0)$ is:

(1) 503

(2) 0

(3) 1

(4) 1006

11. If $y = x^{x^{x \dots \text{to } \infty}}$, then $\frac{dy}{dx} =$

- (1) $\frac{y^2}{x(1-y \log x)}$ (2) $\frac{x^2}{y(1-y \log x)}$
 (3) $\frac{y^2}{1-y \log x}$ (4) $\frac{x^2}{1-y \log x}$

12. If $x^y = e^{x-y}$, then $\frac{dy}{dx} =$

- (1) $\frac{(\log x - 1)}{(1 + \log x)^2}$ (2) $\frac{\log x}{(1 + \log x)^2}$
 (3) $\frac{1}{(1 + \log x)^2}$ (4) $-\frac{1}{(1 + \log x)^2}$

13. If $y = (\tan x)^{(\tan x)^{(\tan x) \dots \text{to } \infty}}$, then the value of $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$ is:

- (1) 1 (2) 2
 (3) 3 (4) 4

14. Find $\frac{dy}{dx}$, when $x = a(t - \sin t)$, $y = a(1 - \cos t)$

- (1) $\sin \frac{t}{2}$ (2) $\cos \frac{t}{2}$
 (3) $\cot \frac{t}{2}$ (4) $\tan \frac{t}{2}$

15. If $x = a\left(t + \frac{1}{t}\right)$ and $y = a\left(t - \frac{1}{t}\right)$, find $\frac{dy}{dx}$

- (1) $\frac{y}{x}$ (2) $\frac{x}{y}$
 (3) $-\frac{y}{x}$ (4) $-\frac{x}{y}$

16. If $x = \sin^{-1}\left(\frac{3\sin t + 4\cos t}{5}\right)$ and

$y = \sin^{-1}\left(\frac{6\cos t + 8\sin t}{10}\right)$, find $\frac{dy}{dx}$

- (1) 1 (2) 0
 (3) -1 (4) $\frac{1}{\sqrt{1-t^2}}$

17. If $x = \log p$ and $y = \frac{1}{p}$, then:

- (1) $\frac{d^2y}{dx^2} - 2p = 0$
 (2) $\frac{d^2y}{dx^2} + y = 0$
 (3) $\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$
 (4) $\frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$

18. If $x = \tan^{-1}\left(\frac{\sqrt{1+\sin t} + \sqrt{1-\sin t}}{\sqrt{1+\sin t} - \sqrt{1-\sin t}}\right)$

and $y = \tan^{-1}\left(\frac{\sqrt{1+t^2}-1}{t}\right)$, then $\frac{dy}{dx} =$

- (1) $\frac{1}{t^2-1}$ (2) t^2+1
 (3) $\frac{1}{t^2+1}$ (4) t^2-1

19. If $y = (1+x)(1+x^2)(1+x^4) \dots (1+x^{2^n})$, then the value of $\frac{dy}{dx}$ at $x = 0$ is

- (1) 0 (2) -1
 (3) 1 (4) 2

20. If $2x^2 - 3xy + y^2 + x + 2y - 1 = 0$, then $\frac{dy}{dx} =$

- (1) $\frac{3y-4x-1}{2y-3x+2}$
 (2) $\frac{3y+4x+1}{2y+3x+2}$
 (3) $\frac{3y-4x+1}{2y-3x-2}$
 (4) $\frac{3y-4x+1}{2y+3x+2}$

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (1)
2. (2)
3. (2)
4. (1)
5. (1)
6. (2)
7. (3)
8. (2)
9. (4)
10. (3)

11. (1)
12. (2)
13. (2)
14. (3)
15. (2)
16. (1)
17. (3)
18. (3)
19. (3)
20. (1)



PW Web/App - <https://smart.link/7wwosivoicgd4>

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